Main points to consider for effluent treatment with Flowform® products

Nitrogen

The smelly forms are ammonia and ammonium. When they are oxidised they become nitrate. When nitrogen is applied as ammonia / ammonium it is relatively inert in an acidic soil but the several forms of nitrifying bacteria oxidise it to nitrate at an unregulated rate so plants are obliged to take up more than the leaves can convert to a variety of amino acids at that point in time. This means that there are nitrates in the leaves that when eaten by animal or human lead towards nitrate poisoning. This is blocking the capacity of haemoglobin to carry oxygen through the system thus causing the system to become anaerobic which is a favorable environment to pathogenic organisms that are connected to many illnesses. The excess nitrate interferes with the capacity of the digestive system to digest. The process of digestion also reduces the nitrate form of nitrogen from nitrate to ammonium that enters the blood stream where the kidneys have to work very hard to remove it again so producing urine that is very high in urea. In the case of cattle this can treble the concentration of urea which gets dumped over a very small area. To be successfully absorbed one needs a deep biologically active soil where plants have deep roots well able to absorb the nitrogen and return it to the growth cycle. The other important thing is for a soil to contain enough calcium to interact with nitrates formed so that is why I suggest that analysis of effluents before and after treatment with Flowform products. Here there have been some tests done after aerobic stirring as is done in the Biodynamic system that show a significant increase in calcium. The other point is that if the nitrogen has already been converted to a nitrate form the application of nitrates to a soil is much more regulated than when converted by soil organisms therefore the likelihood of leaching is reduced.

Phosphor

In New Zealand where superphosphate has been applied over many years there is a buildup of relatively inert tricalcic phosphate in the soil. When nitrate interacts with calcium in this bond phosphorus is released at a rate that is not related to the requirements of the plants so is therefore likely to be lost from an acidic soil. Phosphorus is the other major pollutant in our streams and waterways

Sulphur

Sulphur in the form of sulphide relates to the rotten egg smell. When that is oxidised to sulphate it looses its bad smell. This is further helped if there is adequate calcium, magnesium and potassium for it to interact with so forming a stable agricultural nutrient. It is even better when all the nutrients are held within the humus colloid where the plant accesses them via a biological process as and when needed.

Carbon

The degrading of agricultural soil is an important part of current climate change in several ways. One is displacement of carbon from the soil to the atmosphere. Carbon in the soil is able to hold four times its weight in water so districts with higher carbon soils are less likely to flood and less susceptible to drought. It has also been noted the temperatures are also moderated, Cooler in summer and warmer in winter.

It has been well noted that fish gather round Flowform product out flows so it would be most likely that soil biology would also be invigorated by Figure 8 treated water and effluent.

For farmers who see things in a more mineral analytical concept form it would be good to investigate this aspect as well.